EduAirControl — Software Implementation

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EduAirControl — Software Implementation

> Procedure for preparing, building, configuring, deploying and verifying the system

Servicio Nacional de Aprendizaje · Análisis y Desarrollo de Software – 3145556

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1. Introducción

This document describes how to deploy EduAirControl from the delivered repository. The system integrates a web application, a mobile application, an API Gateway, Spring Boot services, PostgreSQL databases, Redis, RabbitMQ, observability tools and an IoT prototype based on ESP32.

The guide is written for a development, integration, demonstration and deployment-preparation environment. The existence of a Dockerfile, a Kubernetes manifest or an observability configuration means the mechanism is prepared or documented; it does not automatically mean it has been validated for production.

2. Objetivo

Define a reproducible procedure to:

• obtain and organize the source code;

• prepare tools and environment variables;

• build and run the services;

• create and update databases using Liquibase;

• deploy the web frontend and the mobile application;

• integrate the ESP32 prototype with DHT11, KY-038 and MQ-135;

• run health, smoke and acceptance checks;

• publish images via continuous integration;

• prepare a controlled deployment on Kubernetes;

• perform a rollback or recovery when necessary.

3. Alcance y criterios de lectura

Incluido

• Local deployment with docker-compose.yml in microservicios/ea-control-infra.

• Build of api-gateway, ms-security, ms-classroom-management, ms-sensor-management, ms-environment-monitoring, ms-user-management and ms-user-experience.

• Independent PostgreSQL databases per service in the local stack.

• Redis for session, blacklist and rate limiting.

• RabbitMQ for events and asynchronous communication.

• React/Vite web frontend served by Nginx in a Docker image.

• Expo/React Native mobile application with BLE provisioning.

• Deployment manifests for Kubernetes.

• CI for Java services using GitHub Actions.

• Prototype tests with DHT11 and KY-038.

Not confirmed as production

• Final firmware that sends ESP32 readings to the API.

• Metrological calibration of DHT11, KY-038 or MQ-135.

• Final production strategy, cloud provider, domain and certificates.

• Test coverage measured with JaCoCo.

• Complete observability connected to effective metrics of each service.

• Restore of backups tested in an operational environment.

• Final separation of all data and contracts between services.

> Rule: example values, local passwords, example.com domains and demo keys must be replaced before any institutional or production use.

4. Requisitos de implementación

4.1 Herramientas

• | Herramienta | Versión o condición | Uso |

• |---|---|---|

• | Git | 2.40 or higher recommended | Obtain code and manage versions |

• | Docker Desktop | Docker Engine 24 or higher recommended | Local containers |

• | Docker Compose | 2.20 or higher recommended | Orchestrate the stack |

• | Java | JDK 17 | Build and test Spring Boot services |

• | Maven Wrapper | ./mvnw or mvnw.cmd | Compile each service |

• | Node.js y npm | Node 20 recommended for the frontend Docker; compatible with Expo according to the project | Web and mobile frontend |

• | Android Studio | For native Android execution | SDK, emulator and expo run:android |

• | Arduino IDE | For ESP32 testing | Upload test firmware |

• | curl y jq | Available on the machine | Health checks and smoke test |

• | Kubernetes y kubectl | Only for K8s deployment | Apply manifests and inspect pods |

4.2 Requisitos de hardware

• Computer with a recommended minimum of 8 GB of RAM.

• Space for Docker images, dependencies, builds and PostgreSQL volumes.

• ESP32.

• DHT11 for temperature and humidity.

• KY-038 for relative sound level.

• MQ-135 for relative gas/air quality changes.

• Breadboard, jumpers and safe power supply.

• Android/iOS phone or emulator to test the mobile app.

4.3 Puertos

• | Servicio | Puerto publicado |

• |---|---:|

• | API Gateway | 8080 |

• | ms-security | 18081 on host; 8081 inside Docker |

• | ms-classroom-management | 3002 |

• | ms-environment-monitoring | 3003 |

• | ms-sensor-management | 3004 |

• | ms-user-experience | 3006 |

• | ms-user-management | 3007 |

• | PostgreSQL security | 5437 |

• | PostgreSQL classroom | 5433 |

• | PostgreSQL sensors | 5434 |

• | PostgreSQL monitoring | 5435 |

• | PostgreSQL user management | 5438 |

• | PostgreSQL user experience | 5439 |

• | Redis | 6379 |

• | RabbitMQ AMQP | 5672 |

• | RabbitMQ Management | 15672 |

• | Prometheus | 9090 |

• | Grafana | 3001 |

• | OpenTelemetry OTLP | 4317 / 4318 |

• | Frontend web Docker | 80 inside the container; publish on 3000 or 5173 depending on environment |

Before starting, check for port conflicts. On Windows you can use netstat -ano; on Linux/macOS, lsof -i :<puerto>.

5. Organización del código

The delivery contains two main areas:

```
proyecto/
■■■ Docs-ea-control/                      # Documentación de ingeniería
■■■ ea-control-frontend/
■   ■■■ apps/
■       ■■■ web/                          # React + Vite + Nginx
■       ■■■ mobile/                        # Expo + React Native
■■■ microservicios/
■   ■■■ ea-control-api-gateway/
■   ■■■ ea-control-ms-security/
■   ■■■ ea-control-ms-classroom-management/
■   ■■■ ea-control-ms-sensor-management/
■   ■■■ ms-environment-monitoring/
■   ■■■ ea-control-ms-user-management/
■   ■■■ -ms-user-experience/
■   ■■■ ea-control-infra/                   # Compose, K8s, CI y scripts
■■■ manuales/                              # Manuales de usuario, técnico e instalación
```

The central infrastructure is in microservicios/ea-control-infra:

• docker-compose.yml: main stack.

• docker-compose.observability.yml: Prometheus, Grafana and OpenTelemetry Collector.

• scripts/wait-for-services.sh: wait for health endpoints.

• scripts/smoke.sh: registration, login and logout test.

• deploy/k8s/: namespace, apps, databases, Redis, RabbitMQ, secrets and ingress.

• .github/workflows/build-services.yml: tests, build and image publishing.

6. Configuración de variables de entorno

Create environment files from the examples, without versioning real values:

```
cd proyecto/Docs-ea-control
cp .env.example .env

cd ../ea-control-frontend/apps/web
cp .env.example .env

cd ../mobile/EduAirControlMobile/Front-End
cp .env.example .env
```

In the infrastructure stack, Compose accepts variables such as:

• | Variable | Uso local |

• |---|---|

• | GATEWAY_PORT | API Gateway port, default 8080 |

• | CORS_ALLOWED_ORIGINS | Allowed frontend origins |

• | SECURITY_DB, SECURITY_USER, SECURITY_PASSWORD | Security database and credentials |

• | CLASSROOM_DB, SENSOR_DB, MONITORING_DB | Domain databases |

• | JWT_SECRET | Validation for services that still require it |

• | RABBITMQ_USER, RABBITMQ_PASSWORD | Broker credentials |

• | REDIS_PORT | Redis port |

• | REGISTRY, TAG | Image registry and tag |

• | GRAFANA_USER, GRAFANA_PASSWORD | Local Grafana access |

The frontend uses VITE_API_URL=http://localhost:8080. The mobile app accepts EXPO_PUBLIC_API_URL; on a physical phone it must point to the machine LAN IP, for

example http://192.168.1.20:8080.

To generate a temporary local key you can use:

```
openssl rand -base64 32
```

Do not use the credentials guest/guest, change-me-in-vault or default values in production.

7. Implementación local con Docker Compose

7.1 Preparación

• Install Docker Desktop and verify the Linux engine is active.

• Open a terminal at the project root.

• Verify that the service projects are located alongside ea-control-infra.

• Create or review the stack .env file.

• Confirm that production keys, domains and passwords are not in the repository.

7.2 Construcción y arranque

From the infrastructure:

```
cd proyecto/microservicios/ea-control-infra
docker compose build
docker compose up -d
```

To build and start in a single step:

```
docker compose up -d --build
```

The stack builds the Spring Boot services and pulls PostgreSQL 15, Redis 7 and RabbitMQ 3.13 Management. Each application service depends on its primary resources and declares a healthcheck.

Check status:

```
docker compose ps
docker compose logs -f api-gateway
docker compose logs -f ms-security
```

7.3 Espera y comprobación de servicios

The available script waits for the gateway, security, classrooms and sensors health endpoints:

```
chmod +x scripts/wait-for-services.sh scripts/smoke.sh
./scripts/wait-for-services.sh
```

The maximum timeout can be modified:

```
TIMEOUT=300 ./scripts/wait-for-services.sh
```

The smoke test requires jq and should be run only against a development database, because it creates a test user:

```
./scripts/smoke.sh
```

Minimal manual checks:

```
curl -fsS http://localhost:8080/health
curl -fsS http://localhost:8081/api/v1/auth/health
curl -fsS http://localhost:3002/health
curl -fsS http://localhost:3004/health
```

7.4 Stack de observabilidad

To start the main stack and observability:

```
docker compose \
  -f docker-compose.yml \
  -f docker-compose.observability.yml \
  up -d --build
```

Local access:

• Prometheus: http://localhost:9090.

• Grafana: http://localhost:3001.

• OTLP HTTP: http://localhost:4318.

The configuration is prepared in observability/prometheus.yml and observability/otel-collector.yaml. The availability of effective metrics per service must be verified before declaring full observability.

To stop without removing volumes:

```
docker compose down
```

To remove containers and local data — destructive action —:

```
docker compose down -v
```

8. Implementación del frontend web

Ejecución en desarrollo

```
cd proyecto/ea-control-frontend/apps/web
npm install
npm run dev
```

Vite normally serves the frontend at http://localhost:5173. The VITE_API_URL variable must point to the gateway.

Construcción de producción

```
npm run lint
npm run build
npm run preview
```

The Docker image uses two stages: Node 20 builds the app and Nginx serves dist. To build it manually:

```
docker build \
  --build-arg VITE_API_URL=http://localhost:8080 \
  -t eduaircontrol-frontend:local .
```

The Nginx configuration uses try_files so React routes work on page refresh. The image internal port is 80; the external port is defined in the consuming deployment.

9. Implementación de la aplicación móvil

The application is located at ea-control-frontend/apps/mobile/EduAirControlMobile/Front-End.

```
cd proyecto/ea-control-frontend/apps/mobile/EduAirControlMobile/Front-End
npm install
npm start
```

Available commands:

```
npm run android
npm run ios
npm run prebuild
npm run check:android
npm run lint
```

The app uses Expo 54.0.37, React Native 0.81.5 and React 19.1.0. For a physical phone:

```
EXPO_PUBLIC_API_URL=http://192.168.1.X:8080 npx expo start
```

On the Android emulator, 10.0.2.2 represents the host machine. BLE provisioning looks for the device ESP32_Config; Android 12 or higher requires BLUETOOTH_SCAN and BLUETOOTH_CONNECT permissions.

10. Implementación de base de datos y migraciones

The main Compose brings up a PostgreSQL database per domain:

• eduaircontrol_security.

• eduaircontrol_classrooms.

• eduaircontrol_sensors.

• eduaircontrol_monitoring.

• eduaircontrol_user_management.

• eduaircontrol_user_experience.

Each service runs its Liquibase changelogs on startup. Changes are located in each service's src/main/resources/db/changelog. Hibernate uses schema validation; tables should not be created or modified manually to replace migrations.

Recommended procedure:

• Start databases first and wait for their healthy status.

• Start application services.

• Review Liquibase logs.

• Confirm there are no checksum errors or unapplied changes.

• Run a read/write test for each domain.

• Backup before structural changes.

Persistent volumes are security_data, classroom_data, sensor_data, monitoring_data, user_management_data, user_experience_data and rabbitmq_data. Do not run docker compose down -v if data must be preserved.

11. Implementación de la integración IoT

11.1 Componentes

• | Componente | Función |

• |---|---|

• | ESP32 | Controller, reading and Wi-Fi/BLE communication |

• | DHT11 | Air temperature and humidity |

• | KY-038 | Relative sound level via its analog output |

• | MQ-135 | Relative gas/air quality change detection |

• | Protoboard y jumpers | Prototype assembly |

The provided test programs document:

• DHT11 DATA on GPIO 5/D5.

• KY-038 AO on GPIO 34/D34.

• Serial monitor at 115200 baud.

11.2 Secuencia de implementación

• Assemble the ESP32 and sensors on the breadboard.

• Verify common ground, power and continuity.

• Upload and run the DHT11 test first.

• Upload and run the KY-038 test.

• Evaluate the MQ-135 only with proper voltage adaptation.

• Define the single firmware and the real ingestion contract.

• Register the device in the sensor service.

• Provision Wi-Fi via BLE from the mobile app if the firmware supports it.

• Send a test measurement with an individual device credential.

• Verify the measurement is validated, stored and visible on the dashboard.

> The DHT11 and KY-038 have test programs. The final firmware that connects the device to the API, the definitive MQ-135 connection, battery voltage and calibration remain pending confirmation. The KY-038 must not be presented as a calibrated sound level meter and the MQ-135 must not be presented as a precise CO■ ppm meter.

12. Integración continua y construcción de imágenes

The workflow microservicios/ea-control-infra/.github/workflows/build-services.yml runs on push to main, pull requests to main and manual execution.

The current matrix includes six services:

• api-gateway.

• ms-security.

• ms-classroom-management.

• ms-sensor-management.

• ms-user-management.

• ms-environment-monitoring.

Each service configures JDK 17, Maven cache and runs ./mvnw -B verify. On a push to main, the workflow builds and publishes images to GHCR with the tags latest and the commit SHA.

> ms-user-experience exists in Compose and in the code, but does not appear in the current workflow matrix. Before declaring CI complete, that service must be included and verified.

13. Despliegue en Kubernetes

Manifests are located in microservicios/ea-control-infra/deploy/k8s.

Preparación

• Have a Kubernetes cluster and an Nginx Ingress Controller.

• Create the EduAirControl namespace.

• Configure real secrets outside the repository.

• Publish images to a registry accessible by the cluster.

• Replace example tags and the Ingress domain.

• Configure TLS via cert-manager or the institutional mechanism.

Aplicación de manifiestos

```
cd proyecto/microservicios/ea-control-infra
kubectl apply -f deploy/k8s/namespace.yaml
kubectl apply -f deploy/k8s/secrets.example.yaml
kubectl apply -f deploy/k8s/configmap.yaml
kubectl apply -f deploy/k8s/postgres.yaml
kubectl apply -f deploy/k8s/redis.yaml
kubectl apply -f deploy/k8s/rabbitmq.yaml
kubectl apply -f deploy/k8s/apps.yaml
kubectl apply -f deploy/k8s/ingress.yaml
```

Verify the deployment:

```
kubectl get pods -n eduaircontrol
kubectl get svc -n eduaircontrol
kubectl get ingress -n eduaircontrol
kubectl rollout status deployment/api-gateway -n eduaircontrol
kubectl logs deployment/api-gateway -n eduaircontrol
```

The manifests include replicas, readinessProbe, livenessProbe, resource requests and limits. Do not use secrets.example.yaml with real passwords. The K8s configuration must be

reviewed to confirm that all services present in Compose are also deployed, especially ms-user-experience.

14. Verificación y pruebas posteriores

The deployment is considered technically acceptable when at minimum the following are met:

• | Verificación | Criterio |

• |---|---|

• | Contenedores | All required services are running and their health checks respond |

• | Gateway | GET /health responds successfully |

• | Seguridad | Registration, login, renewal and logout work in the test environment |

• | Gateway | Routes are routed to the correct service |

• | Base de datos | Liquibase completes without errors and data persists after restart |

• | Eventos | RabbitMQ receives and delivers events without duplicate effects |

• | Frontend | npm run lint and npm run build complete successfully |

• | Móvil | Expo starts and can communicate with the gateway |

• | IoT | DHT11 and KY-038 deliver test values; ingestion firmware is validated separately |

• | Smoke test | scripts/smoke.sh finishes without error on a development database |

• | CI | All services included in the matrix run mvnw verify |

• | Kubernetes | Pods ready, probes correct and Ingress accessible in a controlled environment |

Coverage, availability, performance and recovery metrics should not be declared met without corresponding evidence.

15. Despliegue, reversión y recuperación

Detención segura

```
docker compose stop
```

Reinicio de un servicio

```
docker compose restart ms-environment-monitoring
```

Reconstrucción después de un cambio

```
docker compose up -d --build ms-environment-monitoring
```

Reversión en Kubernetes

```
kubectl rollout undo deployment/api-gateway -n eduaircontrol
kubectl rollout status deployment/api-gateway -n eduaircontrol
```

Before a database migration:

• stop or isolate write traffic;

• create a backup of the affected databases;

• run the migration in a test environment;

• verify counts, indexes and constraints;

• deploy the compatible code;

• keep a rollback plan.

Rolling back an incompatible migration should not be executed automatically; it must be approved by the technical owner after evaluating data loss or transformation.

16. Seguridad de la implementación

• Do not publish .env files, private keys, tokens, passwords or Kubernetes secrets.

• Replace demonstration RSA keys and rotate them upon any exposure.

• Use CORS_ALLOWED_ORIGINS with explicit origins.

• Keep internal ports on the private network when possible.

• Apply HTTPS for external access.

• Use an individual credential per IoT device, with rotation and revocation.

• Validate JWT signature in each owning service, not only at the gateway.

• Restrict metrics endpoints to the internal network.

• Review Docker images and dependencies before release.

• Do not use the values guest, dev or change-me-in-vault in production.

17. Estado de implementación y pendientes

• | Área | Estado identificado | Acción pendiente |

• |---|---|---|

• | Código de microservicios | Available in the ZIP | Confirm versions and canonical repositories |

• | Docker Compose | Prepared for the full local stack | Run a clean test and document results |

• | PostgreSQL/Liquibase | Changelogs per service available | Test restore and backward migrations |

• | Redis/RabbitMQ | Configured in Compose | Validate resilience and duplicates |

• | Frontend web | Code, Dockerfile and Nginx available | Confirm final external port |

• | Aplicación móvil | Expo, BLE and services available | Validate on physical Android/iOS device |

• | CI | Workflow for six services | Include ms-user-experience and review contracts |

• | Kubernetes | Manifests available | Review images, secrets, domain, TLS and missing services |

• | Observabilidad | Compose and configuration files available | Confirm real metrics, dashboards and alerts |

• | DHT11/KY-038 | Test programs available | Integrate final firmware and validate sensor → API chain |

• | MQ-135 | Component contemplated | Confirm divider, power, calibration and final use |

• | Seguridad | Controls documented; findings open in quality | Close AC-01, AC-02 and AC-03 before production |

18. Entregables y evidencias

To close a deployment it is recommended to keep:

• output of docker compose ps;

• startup logs without Liquibase errors;

• health endpoint responses;

• smoke test report;

• result of npm run lint and npm run build;

• result of ./mvnw verify per service;

• evidence of login and environment query;

• evidence of one DHT11 reading and one KY-038 reading;

• evidence of the API receiving the IoT measurement, when firmware is ready;

• manifests and output of kubectl get pods in K8s;

• backup and test restore record;

• version, commit, date, responsible and environment of each delivery.

19. Glosario

• | Término | Definición |

• |---|---|

• | API Gateway | Entry point that routes requests to services. |

• | CI/CD | Continuous integration and delivery/deployment. |

• | Docker Compose | Tool to run multiple containers in coordination. |

• | Health check | Check that indicates whether a service is available. |

• | Ingress | Kubernetes HTTP/HTTPS entry to services. |

• | Liquibase | Tool for versioning and executing database changes. |

• | RabbitMQ | Messaging broker for asynchronous events. |

• | Redis | In-memory store used for state and cross-cutting capabilities. |

• | Smoke test | Brief test of essential flows after deployment. |

• | BLE | Bluetooth Low Energy used to configure the ESP32. |

• | DHT11 | Low-cost temperature and relative humidity sensor. |

• | KY-038 | Sound detection module with relative output; not a calibrated sound level meter. |

• | MQ-135 | Gas sensor that requires calibration and voltage adaptation. |

• | RTO/RPO | Recovery time objective and maximum acceptable data loss. |

20. Referencias internas

• Docs-ea-control/10-devops/local-setup.md — local preparation.

• Docs-ea-control/10-devops/environments.md — environments and documented pipeline.

• Docs-ea-control/05-architecture/overview.md — architecture and technologies.

• Docs-ea-control/05-architecture/decisions/records/ADR-010-containerization-strategy.md — containerization strategy.

• Docs-ea-control/05-architecture/decisions/records/ADR-011-production-deployment-strategy.md — production deployment strategy.

• microservicios/ea-control-infra/docker-compose.yml — local stack.

• microservicios/ea-control-infra/docker-compose.observability.yml — local observability.

• microservicios/ea-control-infra/scripts/wait-for-services.sh — service wait script.

• microservicios/ea-control-infra/scripts/smoke.sh — smoke test.

• microservicios/ea-control-infra/.github/workflows/build-services.yml — continuous integration.

• microservicios/ea-control-infra/deploy/k8s/ — Kubernetes manifests.

• 08_Calidad_Mejoras — corrective, preventive and improvement actions.

• 04_Propuesta_Tecnica — architecture, risks and technical decisions.